ANCS L1 Rocket Engineering Design Write-Up

Design justification for ANCS L1 high-power certification rocket

Revision: 2026-09-27 | Units: millimeters unless noted

Core design intent: build a safe, manufacturable, cert-oriented Level 1 rocket around a 2.56 in LOC airframe and a 38 mm I-class motor, while keeping custom printed parts within a 300 x 300 x 300 mm printer envelope.

1. Revision Log

Rev	Date	Change
A	2026-09-26	Initial part specification: 280 mm cone, 65 mm shoulder, 3.03 mm fins, 38 mm motor mount.
B	2026-09-27	Printer constraint identified: max build envelope 300 x 300 x 300 mm. Nose revised to 235 mm cone + 60 mm shoulder = 295 mm total.
C	2026-09-27	Fin design justification expanded: swept clipped-delta planform, 4 mm thickness, through-wall tab, airfoil/edge-treatment trade study, and verification plan.

2. Source Basis, Requirements, and Assumptions

This write-up separates source-backed facts, derived calculations, and design judgments. Source-backed facts are referenced with bracketed reference IDs in the text. Calculations are shown from the current geometry. Design judgments are called out as recommendations or open verification items rather than treated as proven facts.

Category	Basis used in design
Certification class	NAR Level 1 high-power certification requires a candidate to build and fly a rocket containing at least one H or I impulse class motor; Level 1 certification allows purchase/use of H and I impulse motors by adult NAR members [R1, R2].
Workflow	ANCS project note directs: design in OpenRocket, CAD needed custom parts, compare CAD to OpenRocket, print/build, then fly on the scheduled flight date [R3].

Simulation basis	OpenRocket displays rocket size, mass, apogee, max velocity, max acceleration, stability, CG, and CP; its documentation defines stability margin as the CG-to-CP distance measured in calibers [R4, R5].
Known purchased airframe	LOC BT-2.56 body tube selected from Apogee listing: 2.560 in ID, 2.630 in OD, 30 in length, 158 g mass [R6].
Known motor family	AeroTech I175WS is a 38 mm I-class motor; currently listed at 333 N-s total impulse and 175 N average thrust [R7, R8].
3D printer constraint	User-provided printer envelope: 300 x 300 x 300 mm. This drives the 295 mm one-piece nose assembly limit.
Material/process	User-provided printed material baseline: ABS, 100% infill. Final print settings remain a manufacturing verification item.

3. Top-Level Requirements

Requirement	Design response	Verification method
L1 certification vehicle	Use one H/I motor; selected baseline motor is AeroTech I175WS, 38 mm.	Confirm motor availability, cert requirements, and final delay with mentor/RSO and OpenRocket.
Fit known airframe	Use LOC BT-2.56 tube with 66.80 mm OD and 65.02 mm ID.	Measure actual received tube ID/OD before final print/slot cuts.
Fit 300 mm printer	Nose assembly revised to 295 mm total length.	Measure CAD bounding box and slicer bounding box.
Maintain stable flight	Three aft through-wall fins with large swept planform; final CG/CP in OpenRocket must meet target margin.	OpenRocket simulation with measured final mass and CG.
Recover safely	30 in compact elliptical parachute, Kevlar shock cord, Nomex blanket, quick links, and metal eye-bolt bulkhead attachment.	Ground-fit recovery bay; verify packing, shock cord routing, and ejection separation.
Buildable by team	Printed nose cone, printed fins, printed centering rings; purchased airframe/motor tube/recovery hardware.	Slicer review, test coupons, dry fit, assembly checklist.

4. Current Baseline Geometry

Subsystem	Current baseline

Motor	AeroTech I175WS-5 target configuration, 38 mm. Public product/source data should be treated as I175WS family data until exact purchased motor/delay is in hand.
Body tube	LOC BT-2.56 / Apogee #11013: ID 65.02 mm, OD 66.80 mm, length 762 mm, listed mass 158 g.
Nose cone body	Von Karman / LD-Haack C=0 profile, cone length 235 mm, base OD 66.80 mm, shoulder OD 64.52 mm, shoulder length 60 mm, total length 295 mm, 2 mm wall.
Nose shoulder fit	Body tube ID 65.02 mm minus shoulder OD 64.52 mm = 0.50 mm diametral clearance = 0.25 mm radial clearance.
Bulkhead ledge	Internal ledge OD 61.00 mm, ID 54.52 mm, thickness 3.00 mm, 10 mm into shoulder from nose/shoulder junction.
Bulkhead disk	Separate printed disk: OD 59.8 mm, thickness 5 mm, 7 mm center hole for 1/4-20 eye bolt hardware.
Motor mount tube	ID 38.74 mm, OD 41.53 mm, length 279.4 mm; aft end flush with body tube aft end. Verify actual tube before CAD freeze.
Centering rings	Two printed rings: OD 64.72 mm, ID 41.83 mm, thickness 6 mm. Aft ring position must be revised/slot-cleared because it conflicts with full-length fin tabs.
Fins	Three printed fins. Exposed planform points: (0,0), (100.08,80), (128,50), (128.27,0). Full-root through-wall tab: 128.27 mm long x 12.64 mm deep. Recommended thickness: 4 mm.
Recovery	30 in compact elliptical parachute, 1/4 in Kevlar shock cord, 10 in Nomex blanket, quick links, and metal eye bolt/nut/washer bulkhead stack.

5. Nose Cone Engineering Justification

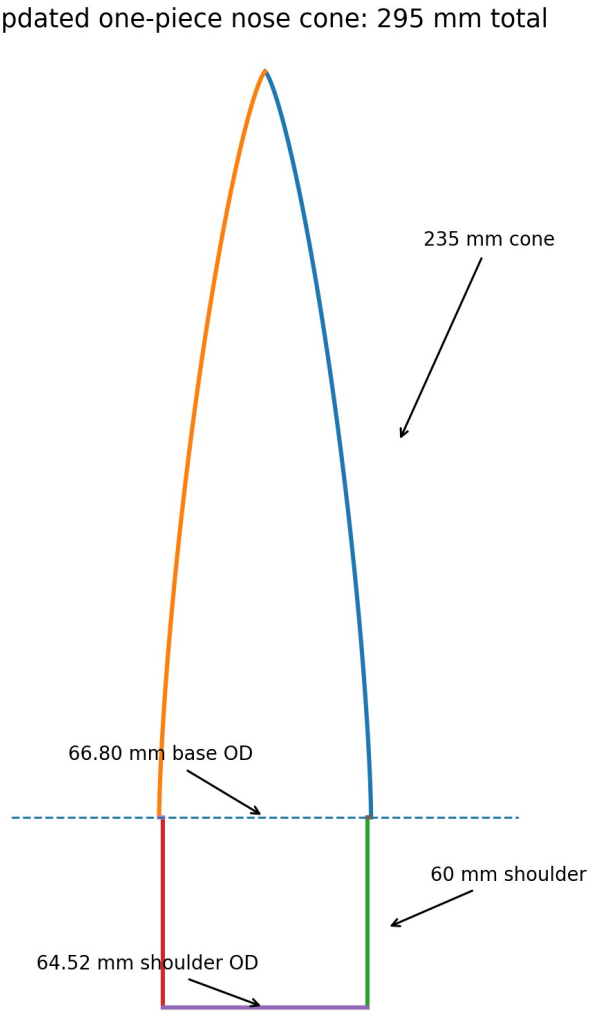


Figure 1. Updated one-piece nose cone geometry sized for a 300 mm printer envelope.

5.1 Shape: Von Karman / LD-Haack C=0

Decision: use a Von Karman / LD-Haack C=0 nose profile rather than a simple cone. The engineering logic is that a Haack-series profile is a standard mathematically derived low-drag nose family, and Von Karman is a common default for high-power rockets where transonic drag may matter. The practical benefit for this build is not that the nose alone guarantees a higher apogee, but that it gives a smooth, credible aerodynamic profile without adding CAD complexity beyond an equation-driven curve [R9, R10].

The updated equation-driven curve is:

Xt = 18.8445*sqrt(t - sin(2*t)/2)

Yt = 117.5*(1 + cos(t))

t1 = 0, t2 = pi

The x coefficient stays fixed because the base radius is unchanged at 33.40 mm. The y coefficient is half the new cone length: 235/2 = 117.5 mm.

5.2 Nose Length and Fineness Ratio

Decision: use 235 mm cone length. The one-piece print must fit a 300 mm tall printer, and the shoulder still needs enough insertion length to align the nose and hold recovery hardware. With a 60 mm shoulder, total nose assembly length is 295 mm, leaving approximately 5 mm of printer height margin.

Candidate	Cone length	Shoulder length	Total	Fineness ratio L/D	Assessment
Original	280 mm	65 mm	345 mm	4.19	Aerodynamically sleek but does not fit 300 mm printer.
Shortened conservative	230 mm	65 mm	295 mm	3.44	Strong shoulder, slightly shorter/stubbier cone.
Selected	235 mm	60 mm	295 mm	3.52	Best current compromise: fits printer, keeps useful shoulder, improves nose length.
Aggressive	240 mm	55 mm	295 mm	3.59	More nose length but shoulder engagement starts becoming less forgiving.

5.3 Shoulder Diameter and Length

Decision: use a 64.52 mm shoulder OD and 60 mm shoulder length. The shoulder OD preserves the intended 0.25 mm radial clearance inside the 65.02 mm tube bore. The shoulder length is shortened from 65 mm to 60 mm only because the printer constraint forces a trade against cone length. A 60 mm shoulder is still 0.90 body diameters of engagement: 60/66.80 = 0.90D.

Risk if shortened further: less engagement length means more sensitivity to print tolerance, tube ovality, wobble, ejection load alignment, and less interior length for the bulkhead stack. Therefore, shoulder lengths below roughly 50 mm are not recommended for this first L1 build unless OpenRocket packaging and a physical fit test force the change.

5.4 Wall Thickness and Bulkhead System

Decision: keep the nose cone hollow with a 2 mm shell. A solid printed ABS nose would move unnecessary mass forward and waste print time/material. A 2 mm wall keeps the external geometry while leaving internal volume for recovery attachment. The recovery harness should not pull directly on a thin printed shell, so a separate disk bulkhead and metal eye bolt are used.

Decision: ledge OD changed to 61.00 mm even though the shoulder bore is about 60.52 mm. This intentionally overlaps the ledge into the shoulder wall by (61.00 - 60.52)/2 = 0.24 mm per side, avoiding a zero-contact feature that may fail to merge cleanly in CAD or print poorly.

Feature	Dimension	Reason
Shoulder bore estimate	60.52 mm	64.52 mm OD - 2x 2.00 mm wall.
Bulkhead disk OD	59.8 mm	Provides 0.72 mm diametral clearance, or 0.36 mm per side, for printed disk insertion.
Bulkhead hole	7.0 mm	Clearance for 1/4 in / 6.35 mm eye-bolt shank and print tolerance.
Ledge ID	54.52 mm	Creates a 3 mm radial shelf from a 60.52 mm bore target.
Ledge thickness	3 mm	Enough printed bearing surface for the disk without consuming much shoulder depth.

6. Motor, Airframe, and Motor Mount Justification

6.1 Motor Selection

Decision: use the AeroTech I175WS family as the baseline. This satisfies the L1 requirement because it is an I-class motor, and NAR Level 1 certification is centered on a successful flight with at least one H or I motor [R1, R2]. Public AeroTech/product sources list I175WS as a 38 mm motor with 333 N-s total impulse and 175 N average thrust [R7, R8].

Important verification item: the exact delay cannot be assumed from the motor name alone. The working spec says I175WS-5, but the actual purchased motor and delay drilling/selection must be confirmed against OpenRocket predicted coast time to apogee before flight.

6.2 Body Tube Selection

Decision: use LOC BT-2.56. The tube is a standard high-power-style airframe size with known published dimensions and mass: 2.560 in ID, 2.630 in OD, 30 in length, and 158 g [R6]. Choosing an off-the-shelf tube reduces schedule risk before the flight date and gives known dimensions for nose shoulder, fin slots, and centering rings.

6.3 38 mm Motor Mount

Decision: use a 38 mm motor mount tube with the aft end flush with the body tube aft end. This supports the selected 38 mm motor family and simplifies motor retention geometry. The current mount tube dimensions are ID 38.74 mm, OD 41.53 mm, length 279.4 mm. Because these dimensions are not yet tied to a received purchased part, they must be verified with calipers before final centering-ring or fin-tab freeze.

6.4 Centering Rings

Decision: use two printed centering rings as the baseline, but do not freeze the aft ring until the fin-tab interference is resolved. The forward ring near the front of the motor tube supports alignment. The aft ring originally placed near 10 mm from the motor tube aft end conflicts with full-depth fin tabs running 0-128.27 mm from the aft end. This is not a minor issue: a ring cannot occupy the same volume as the through-wall tabs unless it is slotted or moved.

Recommended fix: move the aft centering ring just forward of the fin-tab zone, around 135-145 mm from the aft end, or explicitly slot it for the three tabs. Moving it forward is cleaner for manufacturability; slotting it may improve aft support but makes assembly tighter.

7. Recovery System Justification

Decision: use a 30 in compact elliptical Fruity Chutes-style parachute with 1/4 in Kevlar shock cord, Nomex protection, quick links, and a bulkhead-mounted eye bolt. The 30 in compact elliptical size is consistent with an L1-size rocket but must be verified against final mass and desired descent rate. One retailer listing for a 30 in compact elliptical parachute states 3.3 lb at 20 ft/s, which is a useful sizing reference but not a substitute for final mass-based descent calculation [R11].

Decision: use metal hardware for the nose recovery connection. A printed loop or printed eyelet alone is not recommended because recovery loads are impact/ejection events, not static display loads. Washer/printed bulkhead/washer/nylock nut spreads load and avoids pulling directly through a thin 2 mm shell.

8. Fin Design Engineering Justification

> ***The fins are the highest-risk custom printed subsystem because they control static stability, influence drag and apogee, carry aerodynamic bending loads, and force coupled changes to fin slots, centering-ring placement, and motor-mount assembly.***

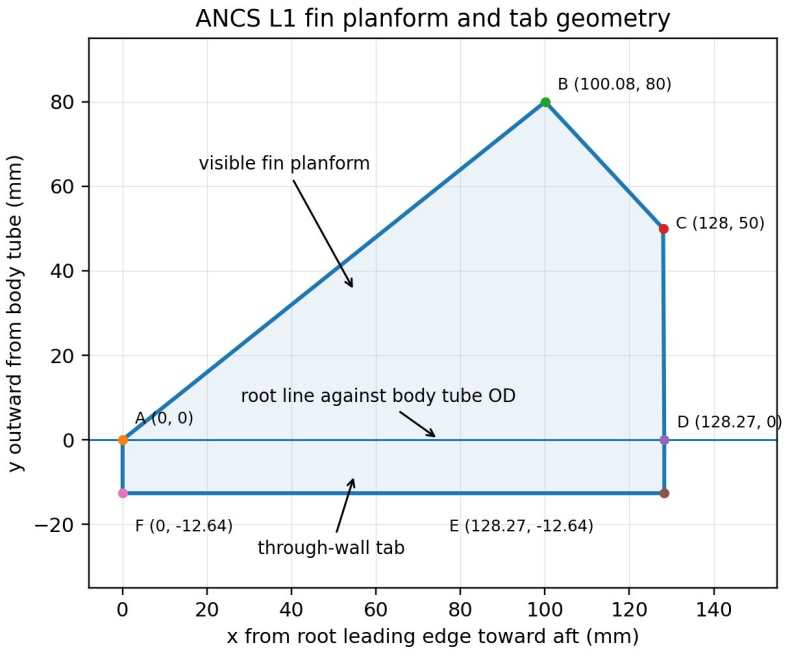


Figure 2. Current fin planform geometry with full-root through-wall tab.

8.1 Number of Fins: Three

Decision: use three fins at 120 degrees. Three fins are the minimum symmetric set that gives roll-balanced stability for a conventional rocket while reducing wetted area, print time, and drag compared with four fins. Four fins would be easier for some visual alignments, but they add 33% more fin count and more drag/weight for the same general stability goal. Three fins are therefore the best baseline for a cert rocket where simplicity, stability, and drag control all matter.

Verification: this is still conditional on OpenRocket. If final CG/CP margin is insufficient, the first correction should be fin geometry or nose/ballast management, not automatically switching to four fins.

8.2 Planform Shape: Swept Clipped-Delta / Triangular Fin

Decision: keep the aggressive triangular/clipped-delta shape rather than converting to a rectangular trapezoid. The current shape provides a large stabilizing area while keeping a swept leading edge and a shorter exposed trailing/outer edge. The sweep also gives the fin the visual style the team wants without making it an arbitrary decorative shape.

Point	Coordinate (mm)	Purpose
A	(0, 0)	Root leading edge at body tube OD.
B	(100.08, 80.00)	Top/span point; sets 80 mm span and swept leading edge.
C	(128.00, 50.00)	Rear upper point; clips the delta shape and avoids a long fragile tip.
D	(128.27, 0)	Root trailing edge; root chord 128.27 mm.

E	(128.27, -12.64)	Aft tab corner; tab reaches toward motor mount.
F	(0, -12.64)	Forward tab corner; full-root tab.

Calculated exposed fin area from A-B-C-D is about 5,825 mm^2. The tab adds about 1,621 mm^2 of printed footprint. The exposed centroid is approximately x = 81.2 mm aft of the root leading edge and y = 28.7 mm outward from the body, useful for sanity-checking CAD and OpenRocket mass/geometry entries.

OpenRocket documentation notes the practical relationship that larger fin area tends to move the center of pressure aft, while CG changes can move the center of mass; this is why the fin dimensions must be compared back to OpenRocket rather than justified only by appearance [R12].

8.3 Fin Span: 80 mm

Decision: keep 80 mm span for the first baseline. The span-to-body-diameter ratio is 80/66.80 = 1.20. That is a large but not insane span for a cert-oriented rocket. The engineering reason is stability margin: the rocket is using a relatively heavy 38 mm motor and a 2.56 in body tube, so aft stabilizing area is useful.

Tradeoff: more span means more drag, more bending load, more flutter risk, and lower apogee. If OpenRocket returns an excessive stability margin, the span can be reduced toward 70-75 mm while preserving the same clipped-delta style. The present decision prioritizes safe, forgiving L1 certification behavior over maximum apogee.

8.4 Fin Root Chord and Tab Length: 128.27 mm

Decision: make the through-wall tab run the full root chord. This is mechanically conservative: aerodynamic loads enter the fin, pass through the printed root, through the body slot, and into the motor mount/fin-can region instead of relying on a surface joint alone. The long root also gives a large bonding/fillet area.

Risk/coupling: full-root tabs force long slots and create the aft centering-ring conflict. That conflict is acceptable only if explicitly resolved in CAD before cutting slots. The body tube slot length remains about 128.3 mm.

8.5 Fin Tab Depth: 12.64 mm

Decision: use 12.64 mm tab depth because it is the radial distance from the body tube OD to the motor mount OD:

(66.80 - 41.53) / 2 = 12.635 mm ≈ 12.64 mm

This means the tab reaches the motor mount OD if the body tube and motor mount are concentric. That is exactly what a through-wall-to-motor-mount fin tab is supposed to do.

8.6 Fin Thickness: 4.00 mm Instead of 3.03 mm

Decision: increase printed fin thickness from 3.03 mm to 4.00 mm. This is one of the most important fin decisions. The mass increase is only proportional to thickness, so 4.00/3.03 = 1.32, or about a 32% increase in fin material. But plate bending stiffness scales approximately with thickness cubed, so the stiffness ratio is:

(4.00 / 3.03)^3 = 2.30

That means the fin becomes roughly 2.3 times stiffer in bending for about 1.32 times the material mass, before accounting for edge shaping. For printed ABS, that stiffness gain is worth it because thin 3 mm fins are more likely to flex, vibrate, or suffer damage during handling and landing.

Manufacturing impact: body tube slot width must be updated. A 3.1 mm slot is no longer acceptable. Recommended slot width is 4.3 mm, giving approximately 0.3 mm diametral clearance over a 4.0 mm printed tab. This fit should be tested on scrap tube or a printed slot gauge before cutting the real body tube.

8.7 Airfoil / Edge Treatment Decision

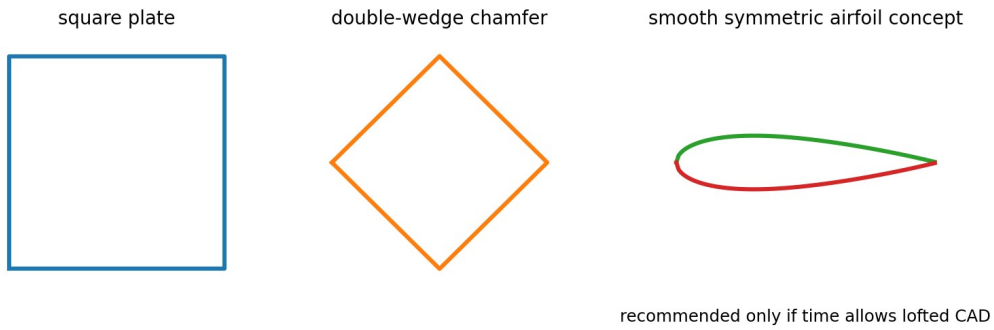


Figure 3. Fin cross-section options: square plate, double-wedge chamfer, and smooth symmetric airfoil concept.

Decision status: the fin should not be described as a true airfoil if it is made only by simple chamfers or fillets. Chamfers and fillets can reduce edge bluntness and make a better rocket fin than a square plate, but a true airfoil requires the full exposed section to follow a smooth symmetric thickness distribution, such as a NACA 00-series style section. NASA/NACA references document symmetric and cambered airfoil families and the calculation of NACA ordinates [R13].

Recommended final approach for this build depends on schedule:

Option	Method	Engineering case	Recommendation
A. Edge-shaped plate	4 mm flat fin; round leading edge; tapered/near-sharp trailing edge; square root/tab.	Fast, strong, printable, and keeps the 4.3 mm body slots. Not a true airfoil.	Use if schedule is tight.
B. True symmetric airfoil	Loft exposed fin as a NACA 0006-ish symmetric section; keep tab 4 mm flat.	Best match to the requested actual airfoil and cleaner pressure recovery. More CAD/slicer risk.	Use if there is time to test print.
C. Meeting chamfers	Chamfer both sides until they nearly meet.	Simple low-drag trailing edge. Sharp leading edge is fragile and not a rounded airfoil.	Trailing edge only.

Engineering recommendation: use a rounded leading edge and a tapered/near-sharp trailing edge at minimum. A rounded leading edge is a common rocket-fin workmanship recommendation because it reduces the penalty of a blunt square leading edge, while a cleaner trailing edge reduces base/pressure drag behind the fin. If the team wants a true “SpaceX-like” or aerospace-style section, do not approximate it with random fillets; CAD the exposed fin as a symmetric NACA-style loft and keep the tab flat. Public SpaceX/Northrop/Lockheed fin profiles should not be treated as available design data unless a real drawing/source is provided.

8.8 Root Fillets

Decision: root fillets are recommended but not yet designed. Printed through-wall fins still need external fillets at the body joint because the body/fin intersection is a stress concentration and drag-producing corner. Fillets increase bonded area, reduce stress concentration, and clean up airflow at the root. The final fillet material and size depend on assembly method: epoxy fillets on ABS/cardboard, printable integrated fillets, or a hybrid.

8.9 Fin Flutter and Speed Risk

Decision status: not cleared. Earlier rough design discussion mentioned approximately 283 m/s max velocity, which is high subsonic/transonic-adjacent. ABS printed fins at 80 mm span need a flutter/strength check before flight. The 4 mm thickness recommendation improves stiffness, but it does not prove flutter safety. Final verification must include OpenRocket max velocity, actual printed material stiffness/print orientation, fin mass, bonding method, and a conservative flutter calculation or mentor review.

8.10 Fin Summary

Decision	Why it is justified	Remaining verification
3 fins	Minimum symmetric stabilizing set; lower drag/weight than 4 fins.	Confirm static margin in OpenRocket.
Clipped-delta planform	Large stabilizing area, swept leading edge, distinctive but still functional geometry.	Confirm CP/stability and drag impact in OpenRocket.
80 mm span	Stability-first L1 baseline.	Reduce to 70-75 mm only if overstable or flutter risk is unacceptable.
Full-root through-wall tab	Transfers fin loads into motor mount/fin-can region.	Resolve aft centering ring interference.
4 mm thickness	About 2.3x bending stiffness vs 3.03 mm for 1.32x material mass.	Update slots to ~4.3 mm and test fit.
Airfoil/edge treatment	Rounded/tapered edges reduce square-plate bluntness; true airfoil requires lofted CAD.	Decide Option A or B before final STL export.
Root fillets	Increase joint area and reduce stress concentration.	Design/apply after fin-can dry fit.

9. Verification and Test Plan

Check	Pass condition	Owner/action
OpenRocket geometry match	OpenRocket body, nose, fin, motor mount, recovery masses match CAD and purchased parts.	Update .ork after CAD freeze.
Static margin	Meets mentor-approved target. OpenRocket guidance: at least 1 caliber subsonic and at least 2 calibers for transonic/supersonic is a documented guideline, not a substitute for RSO/mentor review [R14].	Run loaded motor case and empty motor case checks.
Rail exit speed	Mentor/RSO-approved speed with selected launch rail length and wind conditions.	OpenRocket simulation.
Delay timing	Motor delay matches apogee timing or selected/drilled delay after simulation.	OpenRocket + motor instructions.
Nose shoulder fit	Printed shoulder slides smoothly into measured tube with no binding and no sloppy wobble.	Print short 64.52 mm OD test cylinder before full nose.
Fin slot fit	4 mm tabs fit 4.3 mm slots with adhesive clearance.	Cut test slot or print/cut gauge.
Fin stiffness/flutter	No unresolved mentor concern at predicted max velocity; conservative flutter margin or design reduction.	Run calc / mentor review before flight.
Recovery packing	Chute, shock cord, blanket, quick links fit; ejection separation is not constrained.	Ground pack test and pull test.
Mass and CG override	Final weighed mass and measured CG are entered into OpenRocket before flight.	Weigh complete rocket and measure CG with motor installed.

10. CAD and Export Deliverables

File	Purpose	Export note
ANCS_L1_NoseCone_Body.SLDPRT / .stl	One-piece 235 mm cone + 60 mm shoulder with ledge.	STL in mm, binary, fine resolution, do not scale.
ANCS_L1_NoseCone_Bulkhead.SLDPRT / .stl	Separate recovery bulkhead disk.	Print flat; 7 mm center hole.
ANCS_L1_Fin.SLDPRT / .stl	One fin; print 3 copies.	Finalize airfoil/edge option before STL.
ANCS_L1_CenteringRing_Fwd.SLDPRT / .stl	Forward motor mount centering ring.	ID/OD must match actual tube caliper values.
ANCS_L1_CenteringRing_Aft.SLDPRT / .stl	Aft ring, moved or slotted.	Do not export until tab conflict is resolved.

11. Open Items Before Manufacturing Freeze

- Confirm final fin edge treatment: practical edge-shaped plate vs true lofted symmetric airfoil.
- Resolve aft centering ring versus full-root fin tab conflict.
- Choose motor retention hardware compatible with 38 mm DMS motor and aft geometry.
- Place rail buttons after final internal layout and CG/CP review.
- Run OpenRocket with final measured masses, component CGs, and selected motor/delay.
- Do a fin flutter/strength check for printed ABS at predicted maximum velocity.
- Print fit-test coupons: nose shoulder cylinder, bulkhead disk, fin tab/slot coupon.
- Verify recovery packing volume and ejection separation before flight.

12. References

1. **[R1]** National Association of Rocketry, Level 1 High Power Rocketry Certification Procedures. https://www.nar.org/Level1HPRCertificationProcedures
2. **[R2]** National Association of Rocketry, High Power Rocketry Certification. https://www.nar.org/HPRCertification
3. **[R3]** ANCS project workflow note: design in OpenRocket, CAD custom parts, print/build, fly. Provided project note in current conversation.
4. **[R4]** OpenRocket 23.09 documentation, overview: technical information shown includes mass, apogee, velocity, acceleration, stability, CG, and CP. https://openrocket.readthedocs.io/en/latest/introduction/overview.html
5. **[R5]** OpenRocket technical documentation: stability margin definition as CG-to-CP distance measured in calibers. https://openrocket.sourceforge.net/techdoc.pdf
6. **[R6]** Apogee Rockets, 2.56 in LOC Body Tube / item 11013 listing. https://www.apogeerockets.com/Building_Supplies/Body_Tubes/High_Power_Tubes/2-56in_LOC_Body_Tube
7. **[R7]** AeroTech I175WS-13A 38 mm x 203 mm Single Use DMS listing. https://aerotech-rocketry.com/products/product_34d773c7-e169-6696-041b-b3bdeef1245a

8. **[R8]** ThrustCurve, AeroTech I175WS motor data page. https://www.thrustcurve.org/motors/AeroTech/I175WS/
9. **[R9]** MIT Rocket Team Wiki, How to Choose a Nose Cone. https://wikis.mit.edu/confluence/display/RocketTeam/How%2Bto%3A%2BChoose%2Ba%2BNose%2BCone
10. **[R10]** OpenRocket thesis / technical material discussing Haack-series and Von Karman nose cone shapes. https://openrocket.sourceforge.net/thesis.pdf
11. **[R11]** Rocket Supplies listing for 30 in Compact Elliptical Parachute; states 3.3 lb at 20 ft/s. https://rocket.supplies/proddetail.php?prod=FCCFC-30-S-ST
12. **[R12]** OpenRocket user guide source: larger fin area tends to move CP aft; mass/CG overrides improve simulation accuracy. https://github.com/openrocket/openrocket/blob/unstable/docs/source/user_guide/advanced_rocket_design.rst
13. **[R13]** NASA NTRS, Development of a computer program to obtain ordinates for NACA airfoil families. https://ntrs.nasa.gov/citations/19760003945
14. **[R14]** OpenRocket user guide source: recommended stability margin guideline of not less than 1 caliber subsonic and 2 calibers transonic/supersonic. https://github.com/openrocket/openrocket/blob/unstable/docs/source/user_guide/overrides_and_surface_finish.rst